

QuadStretch: A Forearm-wearable Skin Stretch Display for Immersive VR Experience

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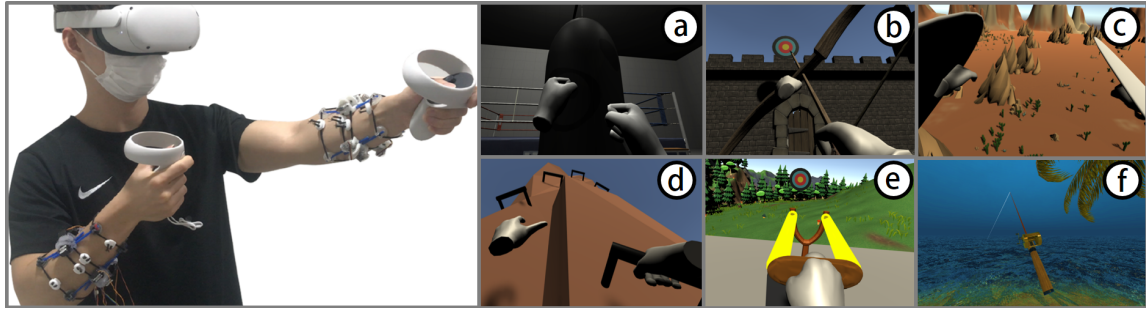


Figure 1: QuadStretch devices worn on both arms and the six application scenarios prepared for this demo. (a) Boxing, (b) Archery, (c) Wings, (d) Climbing, (e) Slingshot, and (f) Fishing.

ABSTRACT

Force feedback is important for immersive VR experience but requires a cumbersome device restricting the use of the hand. An alternative to work around this problem is the use of substituted tactile feedback. QuadStretch is a lightweight and flexible forearm-wearable device for providing substituted skin-stretch feedback to express force on the arm. Consisting of four pairs of tactors, it does not require a ground point and can express the directional sense of force accompanying arm movements. In this demo, we prepared six demo scenarios, Boxing, Archery, Wings, Climbing, Slingshot, and Fishing, to show how the expressive power of QuadStretch can enhance the VR experience.

CCS CONCEPTS

• Human-centered computing → Haptic devices; Virtual reality.

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1 INTRODUCTION

Force feedback to the hand is important for immersive virtual reality (VR) experiences. However, providing force feedback to the hand usually requires a cumbersome mechanical structure and limits the range of hand motion. In addition, force feedback devices restrict bare-hand interaction, which is becoming increasingly important as the current VR systems are equipped with a hand tracker and support hand gesture interaction. To work around the limitations of force feedback devices, researchers are considering a sensory-substitution approach where tactile feedback is provided to a body part that can be physically or mentally related to the point of force feedback [Pezent et al. 2019; Shannon 1976; Wheeler et al. 2010]. Gaitors [Wang et al. 2020] is an example of this approach to provide substituted feedback for force on the leg, and Masque [Wang et al. 2019] is an example of substituted feedback for force on the head. QuadStretch, presented in this demo, is an example of using skin-stretch to provide substituted feedback for force on the arm.

QuadStretch uses skin-stretch feedback because it is expected to match a larger dynamic force range and express a sustained force more naturally than vibrotactile feedback, which is commonly used in tactile feedback devices. To provide skin-stretch feedback, however, a feedback device usually requires a rigid structure with a ground point [Caswell et al. 2012; Tsai et al. 2021; Wang et al. 2019, 2020]. However, this overhead may limit the user adoption of the device in everyday VR applications. QuadStretch uses “paired tactors” to obviate the need for a ground point and uses a flexible structure to fit the shape of the forearm. The paired tactors can apply stretch stimuli without a ground point by counter-acting on each other; however, a downside is that they cannot express directional cues. To overcome this limitation, QuadStretch uses four pairs of tactors around the arm to deliver directional cues

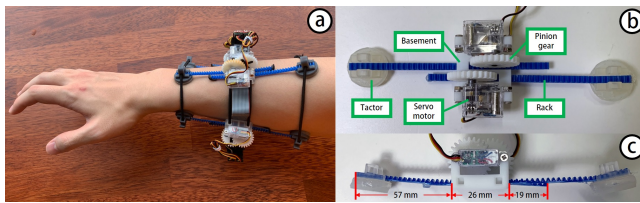


Figure 2: (a) QuadStretch consisting of four stretch modules and (b) Top and (c) side views of the stretch module.

about the force acting on the hand. We demonstrate QuadStretch's dynamic range and rich expressive power, which is continuous and directional, using six VR application scenarios (Figure 1) in this demo.

2 IMPLEMENTATION

As shown in Figure 2, QuadStretch consists of four stretch modules, and each of them consists of two tactors, two rack gears (KHK DR0.8-2000), and two servo motors (HiTec HS-5035 HD). The tactor is a silicone disc with a diameter of 20 mm and a thickness of 3 mm and is attached to the end of the rack. The rack is 3 mm wide and 100 mm long. The pinion gear is 3D printed and has a diameter of approximately 20 mm. The tactor can move by ± 11 mm around the neutral position, but only the range of ± 8.5 mm is used since the participants in a pilot study reported the full range is excessive. The speed of the tactor is about 73 mm/s, and the maximum force of the tactor is about 0.8 kgf. The four stretch modules are placed around the forearm, and connected and secured with elastic strings and cable clips. The eight servo motors are controlled through a driver board (PCA9685, 16-channel 12-bit PWM servo driver) by an Arduino board. The Arduino is connected to the PC through a serial link and receives commands from the Unity VR application. For the demo, we use Oculus Quest 2, which is connected to the PC through Oculus Link so that the PC may control the QuadStretch device while running the demo applications.

3 APPLICATION SCENARIOS

We demonstrate QuadStretch with the six application scenarios described below. The scenarios highlight the QuadStretch's dynamic range (Boxing), continuity (Archery, Wings, and Climbing), and multi-dimensionality (Slingshot and Fishing).

Boxing: The user punches the sandbag and receives an impact through QuadStretch. All four stretch modules of QuadStretch contract up to the maximum travel distance of 8.5 mm and immediately relax. The stretch distance is proportional to the punch speed, and the feedback is completed in up to 0.233 s. This feedback feels like a short shock to the user, and the user can experience how the intensity of the stretch varies according to the punch speed.

Archery: The user pulls the bow and shoots the arrow. The user holds the bow in the left hand and the string in the right hand. When the user pulls the bow string, QuadStretch provides stretch feedback. When the bow is being pulled, all stretch modules in QuadStretch contract in proportion to the distance the arrow is pulled. The user can experience not only the feedback during the action of pulling the bow, but also the feedback caused by the bow tension in the stationary state.

Climbing: There are handles on the boulder, and the user grabs and pulls the handles to climb up. When the user grabs the handle, all modules on the corresponding arm momentarily expand by about 1 cm. As the user pulls the handle, stretch feedback according to the pulling direction and the distance to the handle is provided to express the muscle burden that the user expresses. When the handle is released, the QuadStretch returns to its neutral state.

Wings: This is a relatively surreal scenario. The user has wide wings in place of their hands. The user can fly by flapping the wings. As the user flaps the wings, the air drag force on the wings is expressed as proportional skin stretch. The direction of the drag force is reflected in the skin stretch feedback. For instance, when the user flap down, the modules on the dorsal, ulnar, and radial side of the forearm contract, and the ventral module expands.

Slingshot: The user holds the elastic band of a grounded slingshot by pressing the grip button on the right-hand controller. When the user pulls the band, its elastic force is expressed as proportional skin stretch. The direction of the elastic force is reflected in the skin stretch feedback. For instance, when the rubber band is pulled to the right, the module on the ulnar side of the hand contracts, and the module on the radial side expands. The bullet is fired when the grip button is released.

Fishing: When the fish bites the bait, the fishing line moves, and the force on the fishing rod is expressed as QuadStretch feedback. The fish takes the bait for about a minute and then disappears. The user needs to make a catch gesture by rapidly raising the fishing rod when the fish stops momentarily. As the user freely moves the fishing rod, the QuadStretch feedback changes according to the direction of the fishing line.

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